TerraMind on SmallMinesDS: Experiment Report

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Summary

We trained a shallow UNet decoder on SmallMinesDS 2022 (Sentinel-1 + Sentinel-2, 128 × 128 px) on top of TerraMind v1 tiny, either pretrained and frozen or randomly initialised and trainable, both with D4 augmentation, for N = 1 to 1,491 (all) training images (5 seeds for N ⩽ 250, 3 for N ⩾ 500; 98 runs). Each run trained until its test IoU stopped improving: a check every 5 epochs for N ⩽ 100 and every epoch for N > 100, stop once 20 epochs pass without a gain above 10^{-3} over the best check, learning rate 5e-3 halved at every non-improving check (AdamW, batch min(8, N)); the model at the best check is reported. The frozen pretrained encoder gives the higher IoU at every N ⩾ 5 in every seed, 0.723 vs 0.642 with all images, and with 20 images (0.651) it already exceeds what the trainable encoder reaches with all 1,491. It is also cheaper to train: 2.4 vs 5.5 GFLOP per training sample, and with all images 0.095 vs 0.173 GPU hours (185 vs 417 TFLOP). One effect of the rule is visible in the tables: from N = 250 the check runs every epoch and the learning rate is halved at every non-improving check, so it decays up to five times faster per epoch than at N ⩽ 100; this is why the frozen encoder's IoU at N = 250 (0.653) sits below N = 100 (0.694).

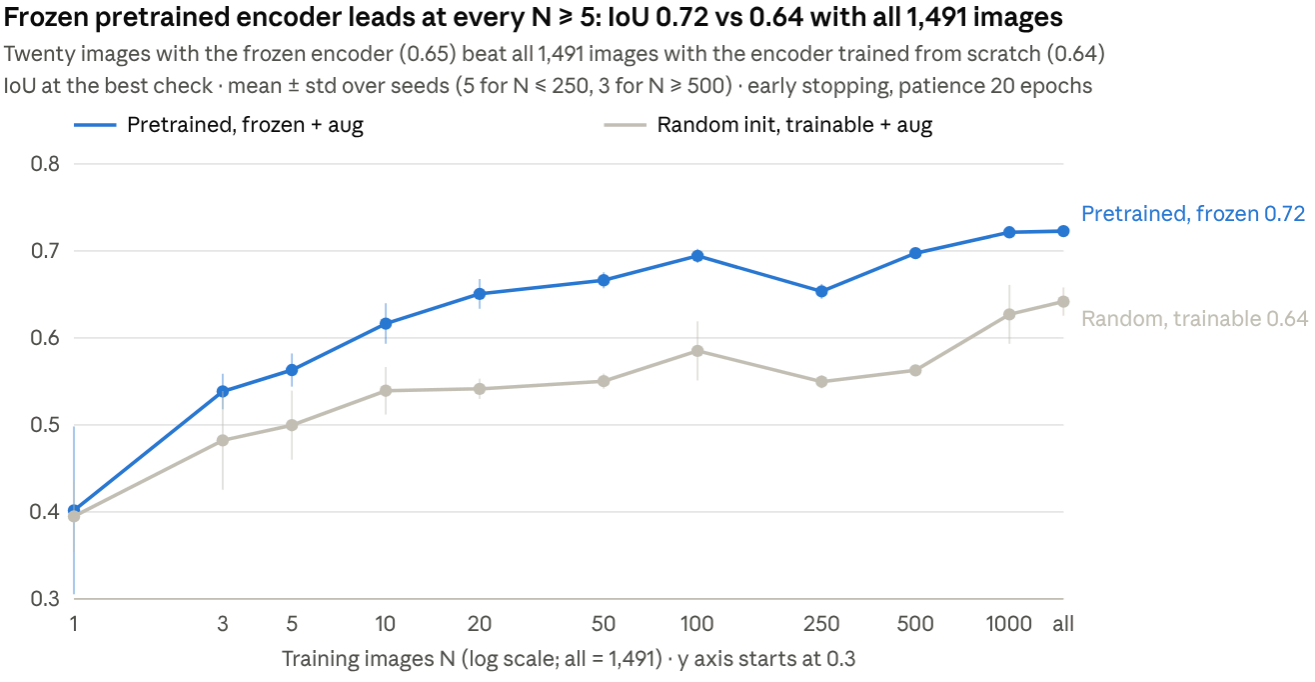


Figure 1. Mine-class IoU on the test set (best check) against the number of training images, mean ± std over seeds

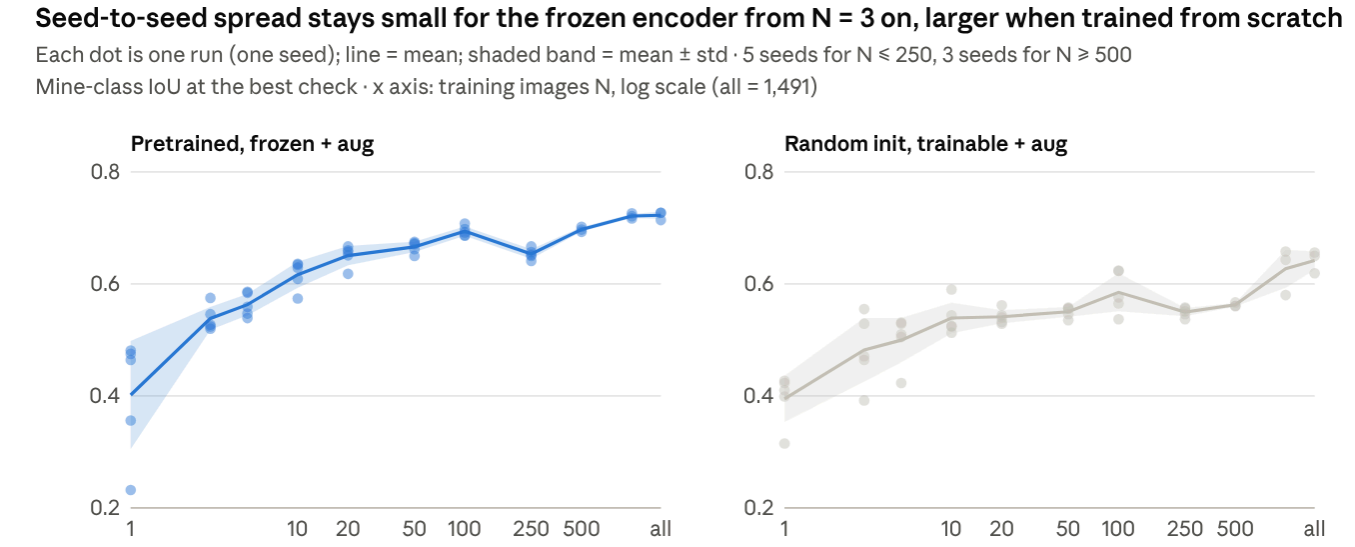


Figure 2. Seed-to-seed variance of IoU: every run as a dot, mean as a line, mean ± std as a shaded band

Table 1. Mine-class IoU on the test set (best check; mean ± std over seeds).

N	seeds	Pretrained, frozen + aug	Random init, trainable + aug
1	5	0.402 ± 0.096	0.395 ± 0.041
3	5	0.539 ± 0.020	0.482 ± 0.057
5	5	0.563 ± 0.019	0.500 ± 0.040
10	5	0.616 ± 0.023	0.539 ± 0.027
20	5	0.651 ± 0.017	0.541 ± 0.012
50	5	0.666 ± 0.009	0.550 ± 0.009
100	5	0.694 ± 0.008	0.585 ± 0.034
250	5	0.653 ± 0.008	0.550 ± 0.008
500	3	0.698 ± 0.004	0.563 ± 0.003
1000	3	0.722 ± 0.004	0.627 ± 0.034
all (1,491)	3	0.723 ± 0.006	0.642 ± 0.016

Table 2. Training cost until early stopping — pretrained, frozen + aug (mean ± std over seeds).

N	best epoch	stop epoch	steps	GPU hours	total TFLOP
1	43.0 ± 28.0	63.0 ± 28.0	63 ± 28	0.003 ± 0.001	15.4 ± 6.8

N	best epoch	stop epoch	steps	GPU hours	total TFLOP
3	39.0 ± 9.2	59.0 ± 9.2	59 ± 9	0.002 ± 0.000	14.7 ± 2.3
5	55.0 ± 8.9	75.0 ± 8.9	75 ± 9	0.003 ± 0.000	19.0 ± 2.3
10	63.0 ± 10.8	83.0 ± 10.8	166 ± 22	0.005 ± 0.001	22.0 ± 2.9
20	76.0 ± 13.9	96.0 ± 13.9	288 ± 42	0.006 ± 0.001	27.7 ± 4.0
50	65.0 ± 17.0	85.0 ± 17.0	595 ± 119	0.010 ± 0.002	30.6 ± 6.1
100	57.0 ± 6.8	77.0 ± 6.8	1001 ± 88	0.015 ± 0.001	36.9 ± 3.2
250	38.4 ± 10.5	58.4 ± 10.5	1869 ± 335	0.032 ± 0.006	105.1 ± 18.9
500	35.0 ± 22.6	55.0 ± 22.6	3465 ± 1421	0.052 ± 0.021	131.6 ± 54.0
1000	25.7 ± 8.5	45.7 ± 8.5	5708 ± 1062	0.077 ± 0.014	163.5 ± 30.4
all (1,491)	19.0 ± 10.0	39.0 ± 10.0	7293 ± 1876	0.095 ± 0.024	185.1 ± 47.6

Table 3. Training cost until early stopping — random init, trainable + aug (mean ± std over seeds).

N	best epoch	stop epoch	steps	GPU hours	total TFLOP
1	34.0 ± 23.5	54.0 ± 23.5	54 ± 24	0.002 ± 0.001	13.3 ± 5.8
3	49.0 ± 28.5	69.0 ± 28.5	69 ± 29	0.003 ± 0.001	17.8 ± 7.4
5	40.0 ± 11.4	60.0 ± 11.4	60 ± 11	0.003 ± 0.001	16.1 ± 3.1
10	25.0 ± 17.6	45.0 ± 17.6	90 ± 35	0.004 ± 0.002	13.4 ± 5.2
20	35.0 ± 27.9	55.0 ± 27.9	165 ± 84	0.005 ± 0.002	19.4 ± 9.8
50	39.0 ± 5.8	59.0 ± 5.8	413 ± 41	0.010 ± 0.001	30.6 ± 3.0
100	49.0 ± 26.3	69.0 ± 26.3	897 ± 342	0.021 ± 0.008	54.9 ± 21.0
250	6.8 ± 2.0	26.8 ± 2.0	858 ± 65	0.022 ± 0.002	69.5 ± 5.3
500	11.3 ± 6.1	31.3 ± 6.1	1974 ± 386	0.046 ± 0.009	124.7 ± 24.4
1000	38.7 ± 26.1	58.7 ± 26.1	7333 ± 3260	0.162 ± 0.069	396.1 ± 176.1
all (1,491)	24.0 ± 13.0	44.0 ± 13.0	8228 ± 2424	0.173 ± 0.051	416.8 ± 122.8

GPU: NVIDIA GeForce RTX 3080 Laptop GPU (8 GB). GPU hours are the wall-clock time of the whole run (model build, training and every test check). FLOPs are counted once per run with torch.utils.flop_counter (2.37 GFLOP per training sample for the frozen encoder, which has no backward pass through the encoder, 5.54 GFLOP for the trainable one; 1.87 GFLOP per evaluated sample for both) and multiplied by the samples processed; the total includes every test check (644 images each), which dominates the cost at small N.

Code, per-run results and full details: github.com/asimarslan11/SmallMinesDS-Experiment-Project.